

## CITYTWIN · REQUIREMENTS

# CityTwin requirements

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## DOCUMENT CONTROL

|             |                                       |         |                |
|-------------|---------------------------------------|---------|----------------|
| Document ID | CTW-REQ-001                           | Version | 0.5            |
| Status      | <b>Draft</b>                          | Date    | 2026-10-01     |
| Author      | Amish Chadha                          | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept         |         |                |
| Repository  | github.com/BoujeeEnjinia1701/citytwin |         |                |
| Source      | docs/03-requirements.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                                                 |
|---------|------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.5     | 2026-10-01 | Amish Chadha | Constructable design (CTW-DDR-003): R13 figures updated; R16 now not met (\$778.00 against \$750)                                                                      |
| 0.4     | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002)                                                                                                                            |
| 0.3     | 2026-09-25 | Amish Chadha | TRL 3 status from CTW-CAL-001; R12 and R17 redefined and R20 added per CTW-DDR-001; R1 restated for the pull paths (D11); reference rates from the sibling TRL 3 notes |
| 0.2     | 2026-09-25 | Amish Chadha | First measurable requirements for TRL 2, with status against the concept estimates                                                                                     |
| 0.1     | 2026-09-25 | Amish Chadha | Initial scaffold                                                                                                                                                       |

# CityTwin requirements

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The decisions in CTW-DDR-001 and CTW-DDR-002 (decided by Amish, 2026-09-25: go with recommendation) set these targets; the constructable design of CTW-DDR-003 (2026-10-01, open for Amish's review) changes no target. R12 covers the pilot on a sheltered site at 0 to 35 °C air (D3, narrowed from 0 to 40 °C by DDR-002), the outdoor range is R20, R17 is a reported figure because the gateway is costed in TwinKit (D1), R10 is restated for the e-paper street kiosk, and R16 is set against the new `budget_usd` of \$750. The status column comes from the TRL 3 calculation note CTW-CAL-001 v0.3. Three requirements are **not met**: R16 (the parts added to make the kiosk buildable take it \$28.00 over budget; an open decision in CTW-DEC-001), R18 (backup in the worst case, until TwinKit fits a larger pack) and R20 (street climate). R1 is **at risk**.

Table 1. Requirements.

| ID  | Requirement                                        | Target                                                                                                                                                                                                                   | Status (CTW-CAL-001)                                                                                                                                            | Verification (TRL 4 or later)              |
|-----|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| R1  | Ingest every node type in the smart city set       | All 9: seven over LoRaWAN through TwinKit; PotholeLog segment files and DockHub hourly aggregates pulled by outbound HTTPS from the operators' servers (D11)                                                             | <b>At risk:</b> six LoRaWAN types met on paper; the pull paths are proposed but not agreed with PotholeLog and DockHub; CrossSafe has no LoRaWAN uplink defined | Interface agreement with each sibling repo |
| R2  | Capacity on one TwinKit gateway                    | 50 nodes or more and 10,000 records per day or more                                                                                                                                                                      | Met on paper: 46 nodes send 5,616 uplinks a day with 0.61 % lost (collisions and downlink blanking); 10,000 a day lose 0.71 %                                   | Packet capture on a live gateway           |
| R3  | Accept only counts and levels                      | Each node type has a payload schema; any field outside it is rejected and logged; no images or audio stored                                                                                                              | Met by design                                                                                                                                                   | Schema review; fuzz test of the decoder    |
| R4  | Suppress small counts in public outputs            | People and vehicle counts below 5 per hour (D6) published as "fewer than 5"                                                                                                                                              | Met by design                                                                                                                                                   | Review of export code                      |
| R5  | No movement traces in public outputs               | PotholeLog published per road segment only, no vehicle tracks or times finer than one day; DockHub per dock per hour, no card IDs                                                                                        | Met by design                                                                                                                                                   | Review of export code                      |
| R6  | Open data export                                   | Hourly CSV and GeoJSON with a documented schema, under CC BY 4.0 (D7), pushed to a public host                                                                                                                           | Met by design: 7,128 rows, 0.86 MB of CSV a day                                                                                                                 | Sample export checked against the schema   |
| R7  | One map for all node types                         | One layer per node type, node health, and an "as of" timestamp on every view                                                                                                                                             | Met by design                                                                                                                                                   | Review of dashboard mock-up                |
| R8  | Data freshness                                     | Map and kiosk show data no older than 30 min in normal operation                                                                                                                                                         | Met on paper, thin margin: 25.9 min worst on the kiosk, 15.6 min on the web map                                                                                 | Timing log on a live system                |
| R9  | Kiosk readable day and night                       | Readable in direct sun and at night at 1.5 m; key text cap height 7 mm or more                                                                                                                                           | Met on paper: 41 px and 16.0 arcmin per capital; about 219 lx from the hood light dimmed to 0.3 W (1,459 lx at its full 2 W)                                    | Mock-up viewed in sun and at night         |
| R10 | Kiosk responds to a button (restated, CTW-DDR-002) | Street kiosk: press acknowledged within 0.5 s by the button's LED ring, which stays lit until the new image is drawn; selected layer shown within 25 s. Indoor LCD variant: selected layer within 5 s (the TRL 2 target) | Met on paper: LED ring in about 30 ms; layer in 19.8 s. The indoor LCD variant is not modeled at TRL 3                                                          | Timed on a bench build                     |
| R11 | Accessible kiosk                                   | Operable parts 380 to 1,220 mm above ground ( <a href="#">ADA forward reach</a> ); overhang from the post no more than 305 mm between 685 and 2,030 mm                                                                   | Met on paper: buttons at 1,120 mm; overhang 200 mm (hood), 186.5 mm                                                                                             | Accessibility audit of the web             |

| ID  | Requirement                                                                     | Target                                                                                                                                                                                                     | Status (CTW-CAL-001)                                                                                                                                                                                                                         | Verification (TRL 4 or later)         |
|-----|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
|     |                                                                                 | ( <a href="#">protruding objects</a> ); same content on a web page that meets WCAG 2.1 AA, linked by QR code and short URL                                                                                 | (cabinet with sun shield); web page not verifiable at TRL 3                                                                                                                                                                                  | page and a site review                |
| R12 | Pilot kiosk survives its climate (redefined, D3; range narrowed by CTW-DDR-002) | Sheltered or indoor site with no direct sun on the window; operate at 0 to 35 °C air; head IP54, cabinet IP55; panel within its 0 to 40 °C rating and the TwinKit pack within its 0 to 45 °C charge window | Met on paper: at 35 °C air the panel is at 35.1 °C (4.9 K margin) and the pack at 39.4 °C (5.6 K margin)                                                                                                                                     | Temperature log in a climate chamber  |
| R13 | Kiosk stands up to wind                                                         | 35 m/s gust with a factor of 1.5 without yield of post or anchors                                                                                                                                          | Met on paper: post 26.6 MPa, 1.94 kN per anchor, with the sun shield                                                                                                                                                                         | Structural design to the local code   |
| R14 | Electrical safety                                                               | Mains only inside the locked cabinet, behind a 30 mA RCBO and a surge protector; only 12 V SELV in the head; all metal earthed                                                                             | Met by design; 12 V peak 26.7 W on a 60 W supply                                                                                                                                                                                             | Inspection by a qualified electrician |
| R15 | Low running power                                                               | 15 W average or less including the TwinKit gateway                                                                                                                                                         | Met on paper: 9.2 W (81 kWh a year), with the hood light dimmed to 0.3 W                                                                                                                                                                     | Energy meter on a bench build         |
| R16 | Kiosk parts cost                                                                | Pilot kiosk (items 1 to 12, 14, 15, 20, 21) within <code>budget_usd</code> of \$750 (raised from \$600, CTW-DDR-002); the outdoor sun shield (item 19) is reported separately                              | <b>Not met:</b> \$778.00 (\$28.00 over) after the parts added for construction (CTW-DDR-003); \$808.00 with the outdoor sun shield. Raising <code>budget_usd</code> to \$800 is proposed, awaiting Amish (CTW-DEC-001)                       | Priced BOM                            |
| R17 | Cost with gateway (redefined, D1)                                               | Kiosk plus TwinKit gateway reported; the gateway is costed and budgeted in the TwinKit repo                                                                                                                | Met on paper (reported): \$1,068.00                                                                                                                                                                                                          | Priced BOM                            |
| R18 | Honest in an outage                                                             | Screen keeps the last image with its timestamp when power fails; gateway rides through 2 h on its backup pack                                                                                              | <b>Not met in the worst case:</b> 2.40 h with a new pack, 1.63 h aged at 0 °C (TWK-CAL-001); a pack of at least 1.84 Ah (now 1.5 Ah) would meet 2 h and has been requested from TwinKit (CTW-DDR-002); the e-paper keeps its image by design | Outage test                           |
| R19 | Secure by default                                                               | No inbound connections from the internet to the gateway; open data pushed and sibling data pulled by outbound connections only; no exposed ports or USB on the kiosk                                       | Met by design                                                                                                                                                                                                                                | Security review                       |
| R20 | Street kiosk climate (new, target after the pilot, D3)                          | Operate from -20 °C to +45 °C air in full sun                                                                                                                                                              | <b>Not met:</b> panel about 79 °C in low sun and 45.1 °C shaded at 45 °C air; a heated, insulated panel bay needs 13.5 W at -20 °C (fitted only where the partner city has frost); behind the sun                                            | Chamber and field tests               |

| ID | Requirement | Target | Status (CTW-CAL-001)                                                                            | Verification (TRL 4 or later) |
|----|-------------|--------|-------------------------------------------------------------------------------------------------|-------------------------------|
|    |             |        | shield the pack stays under 45 °C in full sun up to 37.8 °C air, and is at 52.2 °C at 45 °C air |                               |

## ■ Assumptions

- Node types and reporting rates follow the sibling TRL 3 calculation notes: CurbCount, HeatMap Node, FloodGauge and NoiseMap 96 uplinks a day; AirStreet 288; LoadZone 124 (100 state changes and 24 heartbeats). CrossSafe's hourly activation summary is a CityTwin assumption; CrossSafe has not defined a LoRaWAN uplink.
- Radio loss is calculated (CTW-CAL-001 section A) rather than assumed; the TRL 2 figure of about 3 % was a placeholder.
- The pilot kiosk (R12) stands on a sheltered site; the street range (R20) is the target after the pilot. Any outdoor site uses the cabinet sun shield and a shaded screen orientation as site rules (CTW-DDR-002).
- 35 m/s design gust is an estimate for a first check; the local wind code governs any real installation.